MLflow for Kubernetes: Quickstart Deployment

A comprehensive guide to streamlining machine learning workflows by leveraging Kubernetes orchestration with MLflow tracking and deployment capabilities.

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Why Kubernetes for ML Deployment?

Scalability

Auto-scale resources based on workload demands

High Availability

Self-healing capabilities minimize downtime

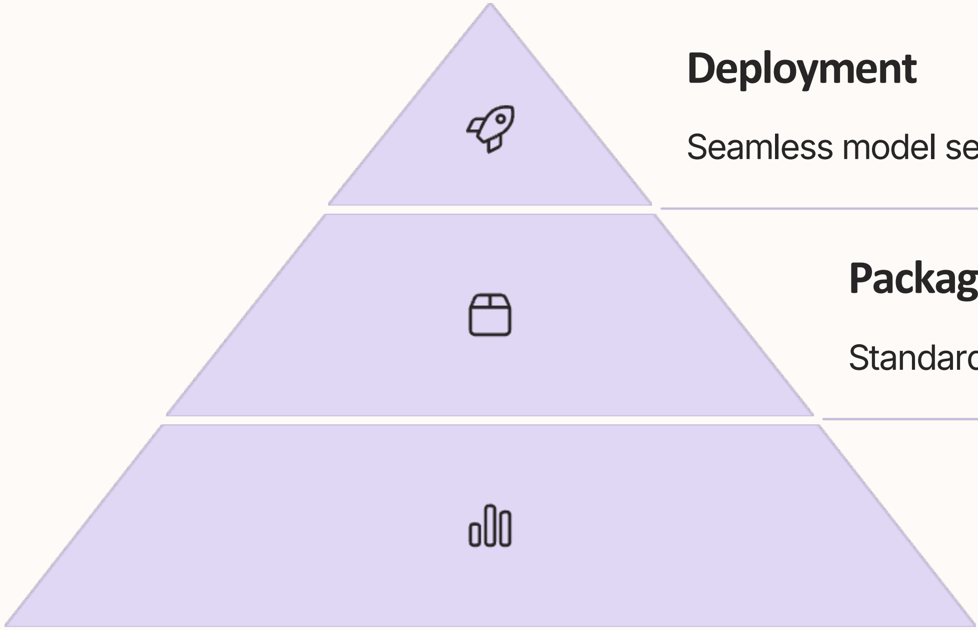
Orchestration

Automated deployment, scaling, and operations

Containerization

Consistent environments across development and production

MLflow's Role in ML Infrastructure



Deployment

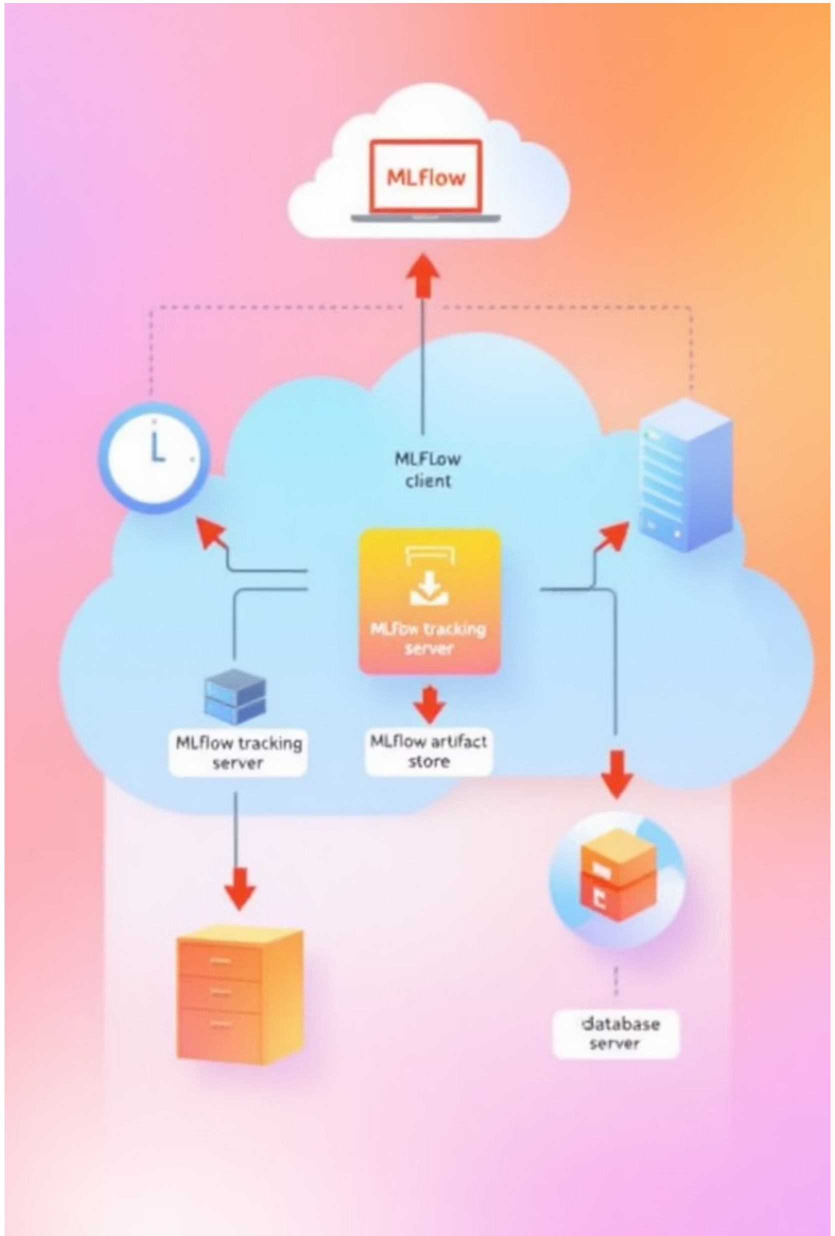
Seamless model serving

Packaging

Standardized model artifacts

Tracking

Experiment metrics and parameters



MLflow Tracking Server Architecture

Experiment Runs

Metrics, params, artifacts logged

Backend Store

SQL database persistence

Artifact Store

S3, GCS, or local storage

K8s Deployment

Scalable tracking service

Kubernetes Resources for MLflow

Deployments

- Tracking server pods
- Model serving instances
- Declarative updates

Services

- Internal communication
- Load balancing
- Stable network identity

ConfigMaps/Secrets

- DB credentials
- S3/GCS access
- Environment configuration

MLflow Model Registry Integration

Version Control

Track model iterations and lineage

Stage Transitions

Staging to production workflow

Access Control

Permissions for model transitions

Deployment Architecture Overview

Model Development

ML experiments tracked with MLflow

Model Packaging

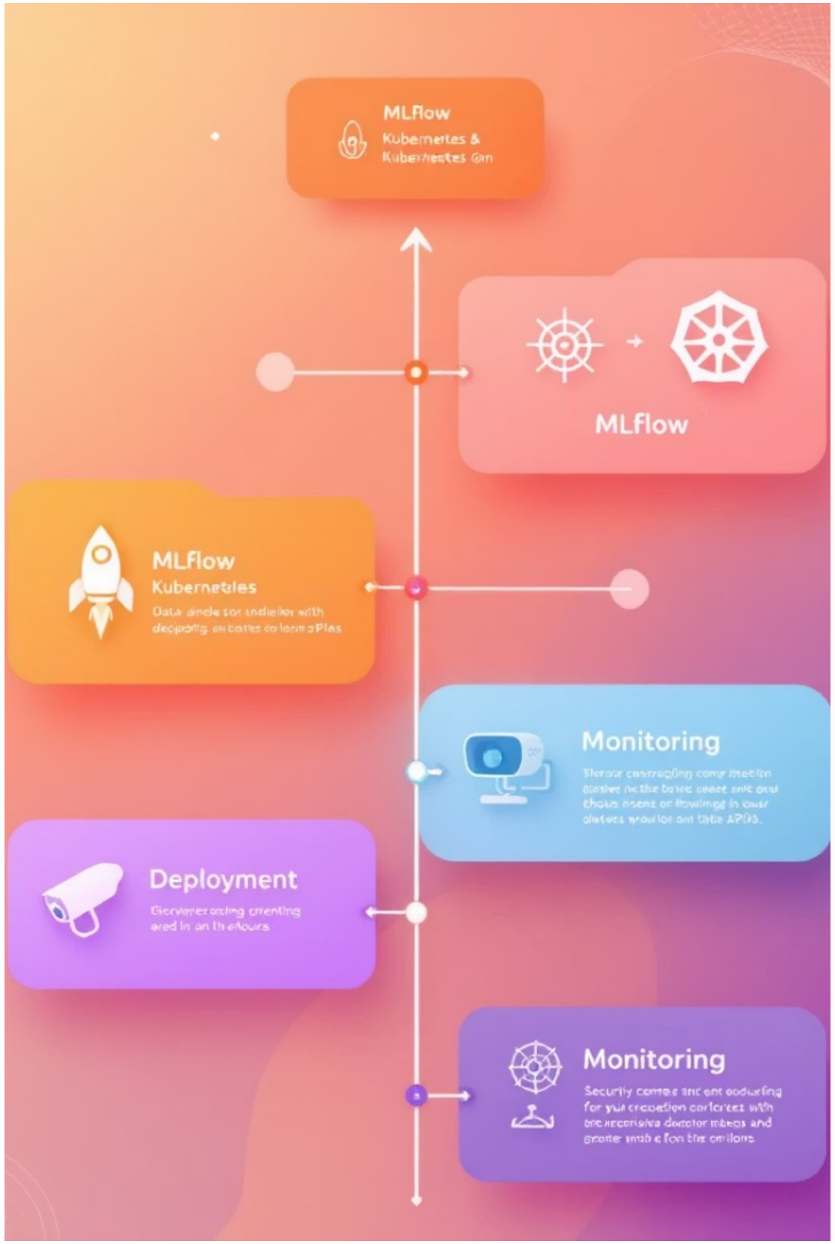
MLflow model format with dependencies

Containerization

Docker images built automatically

K8s Deployment

Scalable inference services



Course Roadmap

Module 1: Environment Setup

K8s cluster configuration and MLflow installation

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Module 2: Tracking Server

Persistent storage and high availability

Module 3: Model Registry

Version control and promotion workflows

Module 4: Inference Deployment

Scalable model serving with monitoring